Calculus I

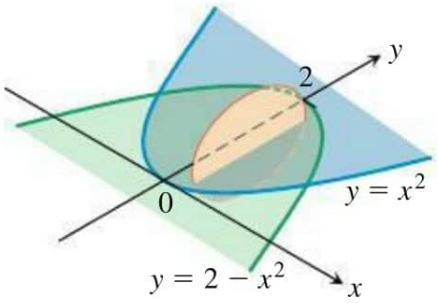
Tutorial #8 (18–22 Nov 2024)

Topics:

- Volume using cross-sections
- Volume using cylindrical shells
- Arc length
- Areas of surfaces

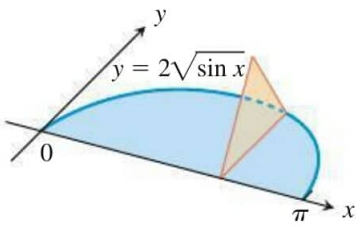
Find the volumes of the solids in Problems 1–2.

Problem 1. The solid lies between planes perpendicular to the x-axis at $x = -1$ and $x = 1$. The cross-sections perpendicular to the x-axis are circular disks whose diameters run from the parabola $y = x^2$ to the parabola $y = 2 - x^2$.



Problem 2. The base of a solid is the region between the curve $y = 2\sqrt{\sin x}$ and the interval $[0, \pi]$ on the x-axis. The cross-sections perpendicular to the x-axis are

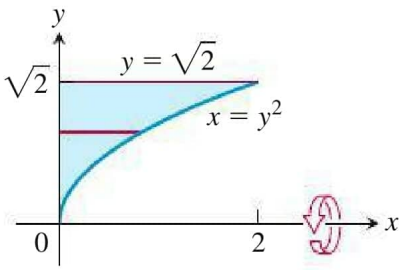
(a) equilateral triangles with bases running from the x-axis to the curve as shown in the accompanying figure.



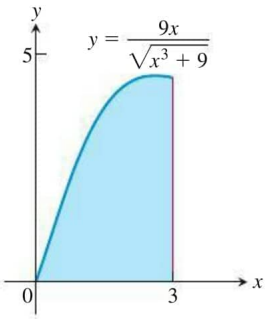
(b) squares with bases running from the x-axis to the curve.

In Problems 3–4, use the shell method to find the volumes of the solids generated by revolving the shaded region about the indicated axis.

Problem 3.



Problem 4. The y-axis



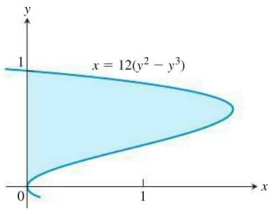
Problem 5. Use the shell method to find the volumes of the solid generated by revolving the shaded region about the indicated axes.

(a) The x-axis.

(b) The line $y = 1$.

(c) The line $y = 8/5$.

(d) The line $y = -2/5$.



Problem 6. Compute the volume of the solid generated by revolving the triangular region bounded by the lines $2y = x + 4, y = x$, and $x = 0$ about

(a) The x-axis using the washer method.

(b) The y-axis using the shell method.

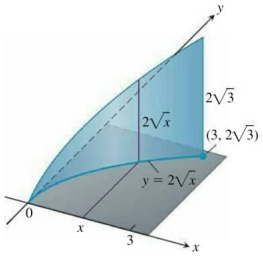
(c) The line $x = 4$ using the shell method.

(d) The line $y = 8$ using the washer method.

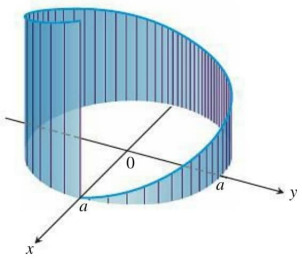
Problem 7. Find the length of the curve $y = \dfrac{x^3}{3} + \dfrac{1}{4x}, \quad 1 \le x \le 3.$

Problem 8. *(The length of an astroid)* The graph of the equation $x^{2/3} + y^{2/3} = 1$ is one of a family of curves called astroid because of their star-like appearance. Find the length of the particular astroid by finding the length of half the first-quadrant portion, $y = (1 - x^{2/3})^{3/2}$, $\sqrt{2}/4 \le x \le 1$, and multiplying by 8.

Problem 9. At points on the curve $y = 2\sqrt{x}$, line segments of length $h = y$ are drawn perpendicular to the xy-plane. (See accompanying figure.) Find the area of the surface formed by these perpendiculars from (0, 0) to $(3, 2\sqrt{3})$.



Problem 10. At points on a circle of radius a, line segments are drawn perpendicular to the plane of the circle, the perpendicular at each point P being of length ks, where s is the length of the arc of the circle measured counterclockwise from $(a, 0)$ to P and k is a positive constant, as shown here. Find the area of the surface formed by the perpendiculars along the arc beginning at $(a, 0)$ and extending once around the circle.



THE END

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Answer to selected problems

1. $\dfrac{16\pi}{15}$
4. 9π
6. (a) 16π (b) $\dfrac{32\pi}{3}$ (c) $\dfrac{512\pi}{3}$ (d) 48π
9. $\dfrac{28}{3}$
10. $2ka^2\pi^2$.